Probability-Valued Subset Sum: Certified Bounds and Learned Guidance

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Companion note III of IV

Abstract. The first note asks when certified coverage can replace explicit search; the second finds no incremental exact subset-sum gain from the tested pairing method. This note changes the question: for fixed weights and a specified random target, what is the probability that some subset reaches it? Witnessed sums and sound exclusions give deterministic bounds on this probability. On 50 held-out 16-item lists, the median lower and upper endpoints from simple rules were 0.8% and 36.0%, while the median exact reachable-target fraction was 24.2%. A learned classifier supplies useful statistical ranking but no per-instance certificate. The bounds and experiments suggest an approximate, costed coverage problem rather than a faster exact decision algorithm.

The probability requires a random experiment

For a fixed list of n positive weights $w = (w_1, \ldots, w_n)$ and an integer target t, let each $x_i \in \{0, 1\}$ indicate whether weight w_i is selected. The decision answer

$$Y(w, t) = \mathbf{1}\left\{ t = \sum_{i=1}^{n} x_i w_i \text{ for some } x \in \{0,1\}^n \right\}$$

is exactly 0 or 1; $\mathbf{1}\{\cdot\}$ is 1 when the condition holds and 0 otherwise. To obtain a probability, let D be a nonempty finite target set, T a random target with distribution μ supported on D, and $\mu(t) = \Pr(T = t)$ its mass at target t. Define the reachable-sum set

$$A(w) = \left\{ \sum_{i=1}^{n} x_i w_i : x \in \{0,1\}^n \right\}.$$

$$q_\mu(w) = \Pr_{T \sim \mu}[Y(w, T) = 1] = \sum_{t \in A(w) \cap D} \mu(t).$$

For a uniform target, $q_\mu(w) = |A(w) \cap D|/|D|$. Thus the new outcome is the chance that a target drawn by a *stated rule* is reachable. Changing D or μ changes the question; the probability is not uncertainty about a fixed target.

Bounds that remain correct for one list

Let $L \subseteq A(w) \cap D$ be distinct sums with explicit subset witnesses. For a positive modulus m, let $R_m(w) = \{a \bmod m : a \in A(w)\}$. Given a finite collection M of moduli, define

$$U = D \cap \bigcap_{m \in M} \{t : t \bmod m \in R_m(w)\}.$$

For any $B \subseteq D$, write $\mu(B) = \sum_{t \in B} \mu(t)$ for its target probability mass.

Proposition 1 (Certified probability interval). For every w, D, μ, L, M defined above,

$$\mu(L) \le q_\mu(w) \le \mu(U).$$

Proof. Every witnessed sum belongs to $A(w)$, giving $L \subseteq A(w) \cap D$. Every attainable sum has an attainable residue modulo every $m \in M$, giving $A(w) \cap D \subseteq U$. Monotonicity of probability mass proves both inequalities. □

Repeated values give another cheap bound. Index the distinct weight values by j, writing v_j for each value and c_j for its multiplicity. One may choose $0, \dots, c_j$ copies of v_j. Hence

$$|A(w)| \le K(w) := \prod_j (c_j + 1).$$

For a uniform target on D, combine the two upper bounds:

$$\frac{|L|}{|D|} \le q_\mu(w) \le \min\left\{\frac{|U|}{|D|}, \frac{K(w)}{|D|}, 1\right\}.$$

The count $K(w)$ is only an upper bound: different count vectors may have the same sum. Exact paired summaries can enlarge L, but their distinct union must be counted, and their construction cost must be charged. More sound exclusions can shrink U.

For a chosen tolerance $\varepsilon > 0$, the interval certifies an additive-ε answer whenever its width is at most ε. Exact q_μ requires zero width. This is the probability analogue of the first note's costed coverage idea: bound quality matters together with the cost of producing it.

A direct check of bound quality

We reused the held-out $n = 16$ lists from the classifier experiment in note II. For each of 50 lists, the target set D was the integer middle 40% of the list's total sum, restricted to targets that passed the experiment's residue tests modulo $8, 10, 16, 32$. Conditional on that filter, targets were drawn uniformly from D. The same residue tests therefore supply no further upper-bound reduction on this particular D.

The lower bound used distinct sums from 512 randomly sampled subsets; the upper bound used $K(w)/|D|$. We enumerated all attainable sums to measure the exact $q_\mu(w)$ *for evaluation only*. Each computed interval contained the exact value.

Table 1: Medians across held-out $n = 16$ lists. Columns are separate medians, not one interval for a representative list.

Family	Lists	Witness lower	Exact q_μ	Count upper
Independent weights	26	0.3%	25.8%	36.9%
Repeated pairs	24	2.2%	24.2%	36.0%
All lists	50	0.8%	24.2%	36.0%

These bounds do not claim a runtime improvement. They show that counting possible selections can sometimes limit the probability without deciding each target, while a small witness budget leaves much of the mass unresolved.

Statistical probabilities and learned guidance

The classifier in note II outputs $\widehat{p}(w, t)$, its estimated YES chance from a feature vector $\phi(w, t)$. For a random list-target pair (W, T) from the experiment's input distribution, the statistical quantity being estimated is $\Pr(Y(W, T) = 1 \mid \phi(W, T) = \phi(w, t))$. This differs from the exact $q_\mu(w)$ for one fixed list: $Y(w, t)$ is fixed once its full instance is known.

The held-out experiment drew four targets per independent list. Its 229 YES answers among 720 targets give an observed rate of 31.8% across the 180 held-out lists. A 95% bootstrap interval resampling whole lists is 26.9%–36.7%. Treating each list's four-target average as one bounded independent observation gives a wider Hoeffding 95% interval

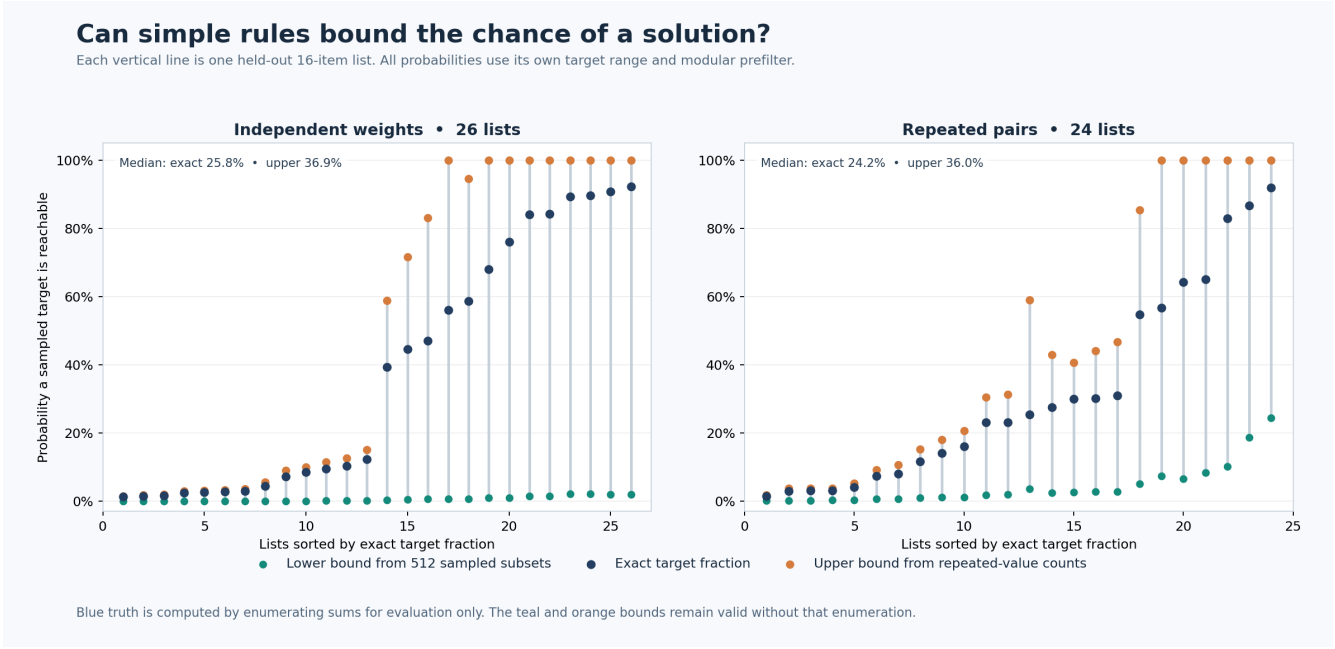


Figure 1: Each vertical segment is a certified interval for one list; the dark point is its exact reachable-target fraction. The low-budget witness bound is often loose.

of 21.7%–41.9% for the mean success rate across that test schedule. On 240 unseen $n = 24$ lists, the observed rate was 25.0% across 960 targets; the list-bootstrap interval was 20.9%–29.2%. These rates depend on the synthetic list and target generators.

For a *fixed* list, draw m independent targets from μ, label them exactly, and let $\widehat{q}$ be the fraction labeled YES. For any $\varepsilon > 0$, the standard sampling bound is

$$\Pr\{|\widehat{q} - q_\mu(w)| > \varepsilon\} \le 2e^{-2m\varepsilon^2}.$$

The probability here is over the m target draws. The exact-label cost remains. Sampling uncertainty intervals are different from Proposition 1's deterministic bounds.

At the classifier threshold $\widehat{p} \le 1\%$, 2 of 43 held-out proposed NO answers were actually YES. The list-bootstrap interval for that conditional error fraction was 0%–10.7%; its zero lower endpoint does not certify zero future risk. A learned score can order which witnesses or certificates to seek, but cannot itself certify a NO decision.

Next question

The useful research target is a cheap sequence of witnessed sets L and sound supersets U whose mass gap closes quickly for a meaningful target distribution. One can measure bound width after a fixed budget, or total work to reach a chosen ε, against direct exact enumeration and sampling with exact labels. The present results motivate this experiment; they do not show a new complexity bound or exact subset-sum speedup.

Note IV pursues a separate extension with one target fixed: if we only need to get close to the target, and know how close, can our framework avoid much of the exponential search? Its output is an actual subset and deterministic bounds on the best possible distance. This changes the accuracy required of coverage rather than introducing a random-target probability.

Its main contribution is a controlled negative timing finding: in the key comparison, pairing gives no consistent additional timing benefit over the same additive approximation rule used sequentially on jointly certified inputs, while completing one additional input within the per-summary state cap. The probability and approximation notes are sibling extensions of Note II and can be read independently after the exact-search test.

Series and reproducibility. Note I: *Certified Coverage of Exponential Search Spaces.* Note II: *Subset Sum as a Stress Test for Certified Coverage.* Note IV: *Does Pairing Add Value to Certified Approximation? A Subset-Sum Experiment.* Scripts, figures, and held-out predictions: `https://github.com/jonland82/exponential-search-spaces`.